

ΑΝΑΦΟΡΑ ΕΡΓΑΣΤΗΡΙΑΚΗΣ ΑΣΚΗΣΗΣ

ΜΑΘΗΜΑ: ΑΡΧΕΣ ΓΛΩΣΣΩΝ ΠΡΟΓΡΑΜΜΑΤΙΣΜΟΥ ΚΑΙ ΜΕΤΑΦΡΑΣΤΩΝ

ΙΟΥΝΙΟΣ 2022

Εργασία των φοιτητών:

ΑΜ	Όνομα	Επίθετο	Έτος	Email επικοινωνίας
1046074	Ιωάννα	Σωτηροπούλου	6	up1046074@upnet.gr
1051205	Γρηγόριος	Αδαμόπουλος	7	gadamopoulos@ceid.upatras.gr
1057777	Ευαγγελία	Ρίνη	6	rini@ceid.upatras.gr

Ημερομηνία υποβολής εργασίας: 19/06/2022

1. α. Συντακτικός ορισμός της BNF γλώσσας που χρησιμοποιήσαμε:

```
jsonfile :          OBRACKETS rSTART CBRACKETS;

rSTART :            rLAST COMMA rACTIVE | rRANGE ;

rRANGE :            rCONTENT COMMA rTOTAL_PAGES COMMA rTOTAL_ELEMENTS COMMA rLAST
COMMA rNUMBER_OF_ELEMENTS COMMA rSORT COMMA rFIRST COMMA rSIZE COMMA rNUMBER;

rLAST :             LAST COLON rBODY_LAST;

rACTIVE :           ACTIVE COLON rBODY_ACTIVE;

rBODY_LAST :        OBRACKETS rGAME_ID COMMA rDRAW_ID COMMA rDRAW_TIME COMMA rSTATUS
COMMA rDRAW_BREAK COMMA rVISUAL_DRAW COMMA rPRICE_POINTS COMMA rWINNING_NUMBERS COMMA
rPRIZECATEGORIES COMMA rWAGER_STATISTICS CBRACKETS;

rBODY_ACTIVE :      OBRACKETS rGAME_ID COMMA rDRAW_ID COMMA rDRAW_TIME COMMA rSTATUS
COMMA rDRAW_BREAK COMMA rVISUAL_DRAW COMMA rPRICE_POINTS COMMA rPRIZECATEGORIES COMMA
rWAGER_STATISTICS CBRACKETS;

rGAME_ID :          GAME_ID COLON INT;

rDRAW_ID :          DRAW_ID COLON INT;

rDRAW_TIME :        DRAW_TIME COLON INT;

rSTATUS :           STATUS COLON rSTATUS_EXT;

rSTATUS_EXT :       RESULTS | ACTIVE;

rDRAW_BREAK :       DRAW_BREAK COLON INT;

rVISUAL_DRAW :      VISUAL_DRAW COLON INT;

rPRICE_POINTS :     PRICE_POINTS COLON OBRACKETS rAMOUNT CBRACKETS;

rAMOUNT :           AMOUNT COLON FLOAT;

rPRIZECATEGORIES :   PRIZE_CATEGORIES COLON OBRACES rPRIZECATEGORIES_EXT CBRACES;

rPRIZECATEGORIES_EXT : rBODY_PC | rPRIZECATEGORIES_EXT COMMA rPRIZECATEGORIES_EXT

rBODY_PC :          OBRACKETS rID COMMA rDIVIDENT COMMA rWINNERS COMMA rDISTRIBUTED
COMMA rJACKPOT COMMA rFIXED COMMA rCATEGORY_TYPE COMMA rGAME_TYPE rMINIMUM_DISTRIBUTED
CBRACKETS;

rWINNING_NUMBERS :   WINNING_NUMBERS COLON rBODY_WN;

rBODY_WN :           OBRACKETS rLIST COMMA rBONUS CBRACKETS;

rINT_ARRAY :        OBRACES rINT_ARRAY_EXT CBRACES;

rINT_ARRAY_EXT :     INT | rINT_ARRAY_EXT COMMA INT;

rLIST:              LIST COLON rINT_ARRAY;

rBONUS :            BONUS COLON rINT_ARRAY;

rID :               ID COLON INT;

rDIVIDENT :          DIVIDENT COLON FLOAT;

rWINNERS :           WINNERS COLON INT;
```

```
rDISTRIBUTED :          DISTRIBUTED COLON FLOAT;

rJACKPOT :              JACKPOT COLON FLOAT;

rFIXED :                FIXED COLON FLOAT;

rCATEGORY_TYPE :       CATEGORY_TYPE COLON INT;

rGAME_TYPE :           GAME_TYPE COLON NORMAL;

rMINIMUM_DISTRIBUTED : COMMA MINIMUM_DISTRIBUTED COLON FLOAT | %empty;

rWAGER_STATISTICS :    WAGER_STATISTICS rBODY_WS;

rBODY_WS :             COLON OBRACKETS rCOLUMNS COMMA rWAGERS COMMA rADD_ON CBRACKETS;

rCOLUMNS :           COLUMNS COLON INT;

rWAGERS :              WAGERS COLON INT;

rADD_ON :              ADD_ON COLON JSON_ARRAY;

rCONTENT :            CONTENT COLON OBRACES rBODY_CONTENT CBRACES;

rBODY_CONTENT :       rBODY_LAST | rBODY_CONTENT COMMA rBODY_CONTENT;

rTOTAL_PAGES :        TOTAL_PAGES COLON INT;

rTOTAL_ELEMENTS :     TOTAL_ELEMENTS COLON INT;

rLAST :               LAST COLON BOOLEAN;

rNUMBER_OF_ELEMENTS : NUMBER_OF_ELEMENTS COLON INT;

rSORT :               SORT COLON rBODY_S;

rBODY_S :             OBRACES OBRACKETS rDIRECTION COMMA rPROPERTY COMMA rIGNORE_CASE
COMMA rNULL_HANDLING COMMA rDESCENDING COMMA rASCENDING CBRACKETS CBRACES;

rDIRECTION :          DIRECTION COLON rDIR;

rDIR :                DESCENDING_SHORT | ASCENDING_SHORT;

rPROPERTY :           PROPERTY COLON STRING;

rIGNORE_CASE :        IGNORE_CASE COLON BOOLEAN;

rNULL_HANDLING :      NULL_HANDLING COLON NATIVE;

rDESCENDING :         DESCENDING COLON BOOLEAN;

rASCENDING :          ASCENDING COLON BOOLEAN;

rFIRST :              FIRST COLON BOOLEAN;

rSIZE :               SIZE COLON INT;

rNUMBER :             NUMBER COLON INT;
```

1. b., 2 Λεξικός και συντακτικός αναλυτής που δημιουργήθηκε, μαζί με την επεκτεταμένη λειτουργία του:

Κώδικας Flex:

```
%option yylineno

%{
#include "myscanner.tab.h"
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

%}

boolean      ("true"|"false")
string       [\""]([\\ \t\ra-zA-Z0-9]|(\\\"))+?[\""]
json_array   \[\]
s_int_f      ({string}|-?{int}|-?{float})
int          [1-9][0-9]*|0
float        {int}\.{int}
ws           [ \t\r]
newline      \n

%%

{int}                {printf("%s", yytext); return INT;}
{json_array}         {printf("%s", yytext); return JSON_ARRAY;}
\:                   {printf("%s", yytext); return COLON;}
\,                   {printf("%s", yytext); return COMMA;}
\.                   {printf("%s", yytext); return DOT;}
\{                   {printf("%s", yytext); return OBRACKETS;}
\}                   {printf("%s", yytext); return CBRACKETS;}
\[                   {printf("%s", yytext); return OBRACES;}
\]                   {printf("%s", yytext); return CBRACES;}
\"last\"             {printf("%s", yytext); return LAST;}
\"active\"           {printf("%s", yytext); return ACTIVE;}
\"gameId\"           {printf("%s", yytext); return GAME_ID;}
\"drawId\"           {printf("%s", yytext); return DRAW_ID;}
\"drawTime\"         {printf("%s", yytext); return DRAW_TIME;}
\"status\"           {printf("%s", yytext); return STATUS;}
\"results\"          {printf("%s", yytext); return RESULTS;}
\"drawBreak\"        {printf("%s", yytext); return DRAW_BREAK;}
\"visualDraw\"       {printf("%s", yytext); return VISUAL_DRAW;}
\"pricePoints\"      {printf("%s", yytext); return PRICE_POINTS;}
\"amount\"           {printf("%s", yytext); return AMOUNT;}
\"winningNumbers\"   {printf("%s", yytext); return WINNING_NUMBERS;}
\"list\"             {printf("%s", yytext); return LIST;}
\"bonus\"            {printf("%s", yytext); return BONUS;}
\"prizeCategories\"  {printf("%s", yytext); return PRIZE_CATEGORIES;}
\"id\"               {printf("%s", yytext); return ID;}
\"divident\"         {printf("%s", yytext); return DIVIDENT;}
\"winners\"          {printf("%s", yytext); return WINNERS;}
\"distributed\"      {printf("%s", yytext); return DISTRIBUTED;}
\"jackpot\"          {printf("%s", yytext); return JACKPOT;}
\"fixed\"            {printf("%s", yytext); return FIXED;}
\"categoryType\"     {printf("%s", yytext); return CATEGORY_TYPE;}
\"gameType\"         {printf("%s", yytext); return GAME_TYPE;}
\"Normal\"           {printf("%s", yytext); return NORMAL;}
\"minimumDistributed\" {printf("%s", yytext); return MINIMUM_DISTRIBUTED;}
\"wagerStatistics\"  {printf("%s", yytext); return WAGER_STATISTICS;}
\"columns\"          {printf("%s", yytext); return COLUMNS;}
\"wagers\"           {printf("%s", yytext); return WAGERS;}
\"addOn\"            {printf("%s", yytext); return ADD_ON;}
```

```

\"content\"                {printf(\"%s\", yytext); return CONTENT;}
\"totalPages\"              {printf(\"%s\", yytext); return TOTAL_PAGES;}
\"totalElements\"          {printf(\"%s\", yytext); return TOTAL_ELEMENTS;}
\"numberOfElements\"       {printf(\"%s\", yytext); return NUMBER_OF_ELEMENTS;}
\"sort\"                    {printf(\"%s\", yytext); return SORT;}
\"direction\"               {printf(\"%s\", yytext); return DIRECTION;}
\"DESC\"                   {printf(\"%s\", yytext); return DESCENDING_SHORT;}
\"ASC\"                    {printf(\"%s\", yytext); return ASCENDING_SHORT;}
\"ignoreCase\"              {printf(\"%s\", yytext); return IGNORE_CASE;}
\"property\"                {printf(\"%s\", yytext); return PROPERTY;}
\"nullHandling\"           {printf(\"%s\", yytext); return NULL_HANDLING;}
\"NATIVE\"                 {printf(\"%s\", yytext); return NATIVE;}
\"descending\"              {printf(\"%s\", yytext); return DESCENDING;}
\"ascending\"               {printf(\"%s\", yytext); return ASCENDING;}
\"first\"                   {printf(\"%s\", yytext); return FIRST;}
\"size\"                    {printf(\"%s\", yytext); return SIZE;}
\"number\"                  {printf(\"%s\", yytext); return NUMBER;}
{boolean}                 {printf(\"%s\", yytext); return BOOLEAN;}
{string}                  {printf(\"%s\", yytext); return STRING;}
{ws}                      printf(\"%s\", yytext);
{float}                   {printf(\"%s\", yytext); return FLOAT;}
{newline}                 {printf(\"%s\", yytext); yylineno++;}
.                          printf(\"\\nUnrecognized character. %s\\n\", yytext);

```

```
%%
```

```

int yywrap(void)
{
    return 1;
}

```

```

void yyerror(char *s)
{
    printf(\"\\nError in line %d, input: %s, near %s\\n\", yylineno, s, yytext);
}

```

Κώδικας Bison:

```
%{
#include <stdio.h>
#include <string.h>
#include <stdlib.h>
void yyerror(char *s);
extern FILE *yyin;
extern FILE *yyout;
extern int yylineno;
// function variables
int error_line = 0; //Γραμμή που βρίσκεται το error
int indent_status = 0;
int counter;
int errors = 0;
int length = 0;
int correctIDs[] = {1100, 1110, 2100, 2101, 5103, 5104, 5106};
int number_of_games = 7;
int found = 0;

%}

%error-verbose

%token COLON
%token COMMA
%token OBRACKETS
%token OBRACES
%token CBRACKETS
%token CBRACES
%token DOT
%token RESULTS
%token AMOUNT
%token NORMAL
%token DESCENDING_SHORT
%token ASCENDING_SHORT
%token NATIVE
%token FLOAT
%token INT_ARRAY
%token JSON_ARRAY
%token LAST
%token GAME_ID
%token DRAW_ID
%token DRAW_TIME
%token STATUS
%token DRAW_BREAK
%token VISUAL_DRAW
%token PRICE_POINTS
%token WINNING_NUMBERS
%token PRIZE_CATEGORIES
%token WAGER_STATISTICS
%token LIST
%token BONUS
%token ACTIVE
%token ID
%token DIVIDENT
%token WINNERS
%token DISTRIBUTED
%token JACKPOT
%token FIXED
%token CATEGORY_TYPE
%token GAME_TYPE
%token MINIMUM_DISTRIBUTED
%token COLUMNS
%token WAGERS
%token ADD_ON
```

```

%token CONTENT
%token TOTAL_PAGES
%token TOTAL_ELEMENTS
%token NUMBER_OF_ELEMENTS
%token SORT
%token DIRECTION
%token PROPERTY
%token IGNORE_CASE
%token NULL_HANDLING
%token DESCENDING
%token ASCENDING
%token FIRST
%token SIZE
%token NUMBER
%token BOOLEAN
%token INT
%token STRING

%%

jsonfile :          OBRACKETS rSTART CBRACKETS;

rSTART :            rLAST COMMA rACTIVE | rRANGE ;

rRANGE :            rCONTENT COMMA rTOTAL_PAGES COMMA rTOTAL_ELEMENTS COMMA rLAST
COMMA rNUMBER_OF_ELEMENTS COMMA rSORT COMMA rFIRST COMMA rSIZE COMMA rNUMBER;

rLAST :              LAST COLON rBODY_LAST;

rACTIVE :            ACTIVE COLON rBODY_ACTIVE;

rBODY_LAST :        OBRACKETS rGAME_ID COMMA rDRAW_ID COMMA rDRAW_TIME COMMA rSTATUS
COMMA rDRAW_BREAK COMMA rVISUAL_DRAW COMMA rPRICE_POINTS COMMA rWINNING_NUMBERS COMMA
rPRIZECATEGORIES COMMA rWAGER_STATISTICS CBRACKETS;

rBODY_ACTIVE :      OBRACKETS rGAME_ID COMMA rDRAW_ID COMMA rDRAW_TIME COMMA rSTATUS
COMMA rDRAW_BREAK COMMA rVISUAL_DRAW COMMA rPRICE_POINTS COMMA rPRIZECATEGORIES COMMA
rWAGER_STATISTICS CBRACKETS;

rGAME_ID :          GAME_ID COLON INT;

rDRAW_ID :          DRAW_ID COLON INT;

rDRAW_TIME :        DRAW_TIME COLON INT;

rSTATUS :           STATUS COLON rSTATUS_EXT;

rSTATUS_EXT :       RESULTS | ACTIVE;

rDRAW_BREAK :       DRAW_BREAK COLON INT;

rVISUAL_DRAW :      VISUAL_DRAW COLON INT;

rPRICE_POINTS :     PRICE_POINTS COLON OBRACKETS rAMOUNT CBRACKETS;

rAMOUNT :           AMOUNT COLON FLOAT;

rPRIZECATEGORIES :   PRIZE_CATEGORIES COLON OBRACES rPRIZECATEGORIES_EXT CBRACES;

rPRIZECATEGORIES_EXT : rBODY_PC | rPRIZECATEGORIES_EXT COMMA rPRIZECATEGORIES_EXT

rBODY_PC :          OBRACKETS rID COMMA rDIVIDENT COMMA rWINNERS COMMA rDISTRIBUTED
COMMA rJACKPOT COMMA rFIXED COMMA rCATEGORY_TYPE COMMA rGAME_TYPE rMINIMUM_DISTRIBUTED
CBRACKETS;

rWINNING_NUMBERS :  WINNING_NUMBERS COLON rBODY_WN;

```

```
rBODY_WN :                OBRACKETS rLIST COMMA rBONUS CBRACKETS;

rINT_ARRAY :              OBRACES rINT_ARRAY_EXT CBRACES;

rINT_ARRAY_EXT :          INT | rINT_ARRAY_EXT COMMA INT;

rLIST:                    LIST COLON rINT_ARRAY;

rBONUS :                  BONUS COLON rINT_ARRAY;

rID :                     ID COLON INT;

rDIVIDENT :               DIVIDENT COLON FLOAT;

rWINNERS :                WINNERS COLON INT;

rDISTRIBUTED :            DISTRIBUTED COLON FLOAT;

rJACKPOT :                JACKPOT COLON FLOAT;

rFIXED :                  FIXED COLON FLOAT;

rCATEGORY_TYPE :          CATEGORY_TYPE COLON INT;

rGAME_TYPE :              GAME_TYPE COLON NORMAL;

rMINIMUM_DISTRIBUTED :    COMMA MINIMUM_DISTRIBUTED COLON FLOAT | %empty;

rWAGER_STATISTICS :       WAGER_STATISTICS rBODY_WS;

rBODY_WS :                COLON OBRACKETS rCOLUMNS COMMA rWAGERS COMMA rADD_ON CBRACKETS;

rCOLUMNS :               COLUMNS COLON INT;

rWAGERS :                 WAGERS COLON INT;

rADD_ON :                 ADD_ON COLON JSON_ARRAY;

rCONTENT :                CONTENT COLON OBRACES rBODY_CONTENT CBRACES;

rBODY_CONTENT :           rBODY_LAST | rBODY_CONTENT COMMA rBODY_CONTENT;

rTOTAL_PAGES :            TOTAL_PAGES COLON INT;

rTOTAL_ELEMENTS :         TOTAL_ELEMENTS COLON INT;

rLAST :                   LAST COLON BOOLEAN;

rNUMBER_OF_ELEMENTS :     NUMBER_OF_ELEMENTS COLON INT;

rSORT :                   SORT COLON rBODY_S;

rBODY_S :                 OBRACES OBRACKETS rDIRECTION COMMA rPROPERTY COMMA rIGNORE_CASE
COMMA rNULL_HANDLING COMMA rDESCENDING COMMA rASCENDING CBRACKETS CBRACES;

rDIRECTION :              DIRECTION COLON rDIR;

rDIR :                     DESCENDING_SHORT | ASCENDING_SHORT;

rPROPERTY :               PROPERTY COLON STRING;

rIGNORE_CASE :            IGNORE_CASE COLON BOOLEAN;

rNULL_HANDLING :          NULL_HANDLING COLON NATIVE;

rDESCENDING :             DESCENDING COLON BOOLEAN;
```

```

rASCENDING :          ASCENDING COLON BOOLEAN;

rFIRST :              FIRST COLON BOOLEAN;

rSIZE :              SIZE COLON INT;

rNUMBER :            NUMBER COLON INT;

%%

void check_gameID(int gID){
    /*Check if the gameID is one of the available ones*/
    for(counter = 0; counter<number_of_games; counter++){
        if(gID == correctIDs[counter])
            found = 1;
    }
    if(found = 0){
        errors++;
        error_line = yylineno;
        printf("Error in line %d: the gameID is not available!", &error_line);
        exit(EXIT_FAILURE);
    }
}

void check_prizeCategories(char* pC){
    /*Check if the number of objects within element winningNumbers is 8*/
    // validate zero in paronomasti??
    length = sizeof(pC) / sizeof(pC[0]);
    if(length != 8){
        errors++;
        error_line = yylineno;
        printf("Error in line %d: the element prizeCategories does not have exactly 8
objects!", &error_line);
        exit(EXIT_FAILURE);
    }
}

//To be checked: can we access the list of winningNumbers just by passing a pointer

void check_winningNumbers(char* wN){
    /*Check if the number of objects within element winningNumbers is 5*/
    // pos tha vroume to megethos tou pinaka?
    length = sizeof(wN) / sizeof(wN[0]);
    if(length != 5){
        errors++;
        error_line = yylineno;
        printf("Error in line %d: the element winningNumbers does not have exactly 5
objects!", &error_line);
        exit(EXIT_FAILURE);
    }

    /*Check if the objects within element winningNumbers are in the frame 1-45*/
    for(counter = 0; counter<5; counter++){
        if(wN[counter]<1 || wN[counter]>45){
            errors++;
            error_line = yylineno;
            printf("Error in line %d: the element winningNumbers include objects out of
the frame 1-45!", &error_line);
            exit(EXIT_FAILURE);
        }
    }
}

int main (int argc, char **argv) {
    FILE *jfile = fopen(argv[1], "r");
    if (jfile == NULL)

```

```
{
    printf( "Failed to read input file." ) ;
    exit(EXIT_FAILURE);
}
else {
    printf( "Reading input file...\n" ) ;
    yyin = jfile;
    int token;
    for(;;){
        token = yyparse();
        if (token == 0){
            if(errors==0) {
                printf("\nNo errors identified. Input file is acceptable!\n");
                break;
            } else {
                printf("\nErrors encountered. Error count: %d\n", errors);
                break;
            }
        } else {
            break;
        }
    }
    fclose(jfile);
    printf( "File closed successfully.\n");
    return 0;
}
}
```

Παραδοχές:

Θεωρήσαμε τα παρακάτω:

- Αν και στη λειτουργία του JSON δεν έχει σημασία η σειρά των στοιχείων, από την εκφώνηση της άσκησης και τα δοσμένα αρχεία θεωρήσαμε ότι πρέπει να εμφανίζονται όλα τα στοιχεία εντός των κανόνων της γραμματικής.
- Θέσαμε το στοιχείο `add_on` ως μηδενικό array

Βοηθητικές συναρτήσεις:

Υλοποιήσαμε τις βοηθητικές συναρτήσεις που ζητήθηκαν, αλλά λόγω χρόνου δεν καταφέραμε να τις ενσωματώσουμε στον κώδικα της `main`.

Screenshots λειτουργίας του λεξικού και συντακτικού αναλυτή:

Ο αναλυτής που υλοποιήθηκε επεξεργάστηκε επιτυχώς τα δοσμένα αρχεία `last_result.json` και `range_result.json`. Παρατίθεται τα σχετικά screenshots:

last_result.json

Προκειμένου να χρησιμοποιήσουμε τον αναλυτή, καλέσαμε τις παρακάτω εντολές:

```
bison -d myscanner.y
```

```
flex myscanner.l
```

```
gcc myscanner.tab.c lex.yy.c -lfl
```

```
./a.exe last_result.json
```

```
griga@DESKTOP-JNT5AK8 ~/compilers
$ bison -d myscanner.y
myscanner.y:20.1-14: warning: deprecated directive: '%error-verbose', use '%define parse.error verbose' [-Wdeprecated]
 20 | %error-verbose
    | ^~~~~~
    | %define parse.error verbose
myscanner.y: warning: 2 shift/reduce conflicts [-Wconflicts-sr]
myscanner.y: note: rerun with option '-Wcounterexamples' to generate conflict counterexamples
myscanner.y: warning: fix-its can be applied. Rerun with option '--update'. [-Wother]

griga@DESKTOP-JNT5AK8 ~/compilers
$ flex myscanner.l

griga@DESKTOP-JNT5AK8 ~/compilers
$ gcc myscanner.tab.c lex.yy.c -lfl
myscanner.tab.c: In function 'yyparse':
myscanner.tab.c:1557:16: warning: implicit declaration of function 'yylex' [-Wimplicit-function-declaration]
1557 |         yychar = yylex ();
     |                ^~~~~~
myscanner.tab.c:1719:18: warning: passing argument 1 of 'yyerror' discards 'const' qualifier from pointer target type [-Wdiscarded-qualifiers]
1719 |         yyerror (yymsgp);
     |                ^~~~~~
myscanner.y:5:20: note: expected 'char *' but argument is of type 'const char *'
 5 | void yyerror(char *s);
   |                ~~~~~^
```

```

        "distributed": 0.0,
        "jackpot": 0.0,
        "fixed": 50.0,
        "categoryType": 1,
        "gameType": "Normal"
    },
    {
        "id": 6,
        "divident": 0.0,
        "winners": 0,
        "distributed": 0.0,
        "jackpot": 0.0,
        "fixed": 2.0,
        "categoryType": 1,
        "gameType": "Normal"
    },
    {
        "id": 7,
        "divident": 0.0,
        "winners": 0,
        "distributed": 0.0,
        "jackpot": 0.0,
        "fixed": 2.0,
        "categoryType": 1,
        "gameType": "Normal"
    },
    {
        "id": 8,
        "divident": 0.0,
        "winners": 0,
        "distributed": 0.0,
        "jackpot": 0.0,
        "fixed": 1.5,
        "categoryType": 1,
        "gameType": "Normal"
    }
],
"wagerStatistics": {
    "columns": 0,
    "wagers": 0,
    "addOn": []
}
}
}
No errors identified. Input file is acceptable!
File closed successfully.

```

range_result.json

Προκειμένου να χρησιμοποιήσουμε τον αναλυτή, καλέσαμε τις παρακάτω εντολές:

```
bison -d myscanner.y
```

```
flex myscanner.l
```

```
gcc myscanner.tab.c lex.yy.c -lfl
```

```
./a.exe range_result.json
```

```

        "divident":2.0,
        "winners":8441,
        "distributed":16882.0,
        "jackpot":0.0,
        "fixed":2.0,
        "categoryType":1,
        "gameType":"Normal"
    },
    {
        "id":8,
        "divident":1.5,
        "winners":40987,
        "distributed":61480.5,
        "jackpot":0.0,
        "fixed":1.5,
        "categoryType":1,
        "gameType":"Normal"
    }
],
"wagerStatistics":{
    "columns":1540071,
    "wagers":283493,
    "addon":[]
}
},
],
"totalPages":1,
"totalElements":4,
"last":true,
"numberOfElements":7,
"sort":[
    {
        "direction":"DESC",
        "property":"id.drawId",
        "ignoreCase":false,
        "nullHandling":"NATIVE",
        "descending":true,
        "ascending":false
    }
],
"first":true,
"size":10,
"number":0
}

```

No errors identified. Input file is acceptable!
File closed successfully.

